Implementation change report

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Initial prompt

Implement plans discussed above

Follow-up prompt

Do it now and give me a PDF report with timestamp, date and initial prompt about what was done and what changes happened

Changes

Added a resume production field requiring the original application request, production TPR and checkpoint. Resume uses mdrun -cpi with -append without rebuilding the TPR. Added MDAnalysis input preflight: topology/trajectory loading, group bounds or known selections, finite time intervals and actual coverage. Equal atom counts do not prove ordering; the warning is retained in input-validation.json.

Windows to WSL

Added a command adapter preserving an explicit WSL distribution and translating paths with wslpath as separate arguments. The Python worker runs inside WSL so validation, GROMACS and output parsing share Linux paths. Install requirements inside that distribution separately.

Verification and limits

Compilation was checked. Live WSL probing returned access denied, so no real GROMACS or checkpoint-resume run was verified. Resume currently covers production only. The original checkpoint is checked for existence; GROMACS performs consistency and append validation. Corrupt frames and topology ordering are not automatically repaired. Windows termination does not yet guarantee graceful checkpoint creation inside WSL. Full supplied backend tests require unavailable Python dependencies.

Files

workflow_ui.py, workflow_worker.py, workflow_validation.py, wsl_execution.py, requirements.txt and tests/test_workflow_execution.py. Changes are local and uncommitted.